

VENKATESH MAHALINGAM

Python Developer | Backend Engineer | Full Stack Developer

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PROFESSIONAL SUMMARY

Results-driven Python Developer and Backend Engineer with 3+ years of experience designing, developing, and optimizing scalable web applications. Proficient in Python, Django, Flask, and REST API development with a proven track record of improving system performance by up to 40% through query optimization, caching strategies, and architectural refactoring. Experienced in JWT authentication, OAuth2, Django REST framework, and RBAC-based security implementations adhering to OWASP best practices. Hands-on exposure to AWS (EC2, S3, RDS), Docker, and CI/CD pipelines. Adept at collaborating in Agile/Scrum environments to deliver high-quality, maintainable software across full-stack projects.

SKILLS SUMMARY

Languages: Python (OOP & MVT), JavaScript (ES6+)

Frameworks: Django, Next.js, Bootstrap, Tailwind CSS

Database: MySQL, PostgreSQL, SQLite, MongoDB, GraphQL

Libraries: React.js, jQuery, Ajax, Pandas, NumPy, Matplotlib, Scikit-Learn

DevOps & Cloud: Linux, Docker, Kubernetes, AWS (EC2, S3, IAM, RDS), Jenkins, Terraform, Prometheus, Grafana

API Technologies: REST APIs, JWT Authentication, OAuth2, GraphQL

Version Control: Git, GitHub

Engineering: OOP, MVT Architecture, SOLID Principles, Design Patterns, SDLC, Agile/Scrum, Code Review

AI Tools: Claude AI, GitHub Copilot

WORK EXPERIENCE

Fortigrd ICT India Pvt Ltd | [LINK](#)

Python Developer | Backend Engineer | Full Stack Developer

February 2023 –Present

- Engineered scalable, high-performance Python and Django applications, improving overall system efficiency by 30%.
- Designed and consumed RESTful APIs to integrate microservices and third-party platforms, reducing inter-service communication latency and enhancing system interoperability.
- Refactored legacy Python codebases, eliminating 40% of technical debt while improving maintainability, code readability, and security posture, through optimized SQL queries, indexing strategies, and Redis caching.
- Implemented JWT authentication, OAuth2, and RBAC-based authorization across multiple projects, enforcing OWASP security best practices for data protection.
- Collaborated with cross-functional teams including UI/UX designers and DevOps engineers within Agile/Scrum sprint cycles to deliver production-ready web solutions on schedule.

PROJECTS

LMS – Learning Management System | [LINK](#)

January 2024 –Present

Tech Stack: Django, Bootstrap, jQuery, MySQL

- Contributed to the development of a live Learning Management System (LMS), enabling students to access online courses and educational resources with a seamless digital experience.
- Implemented role-based access control (RBAC) to manage tiered permissions for students, instructors, and administrators, ensuring data integrity and access governance.
- Designed responsive UI with Bootstrap and jQuery, ensuring full cross-device compatibility and smooth user experience across desktop and mobile platforms.
- Enhanced platform security with CSRF protection, input validation, and bcrypt-encrypted user credentials following OWASP security guidelines.

IBUYKENYA – eCommerce Platform | [LINK](#)

May 2024 –Present

Tech Stack: Next.js, FAST API, MySQL

- Architecting and developing a scalable e-commerce platform supporting seamless online shopping, product catalog management, and order processing for the Kenyan market.

- Designed and optimized product listing, order management, and payment gateway integration modules to deliver a high-performance user experience.
- Developed RESTful APIs enabling third-party service integration for payment processors, logistics providers, and user authentication systems.
- Optimized MySQL database with indexing and query caching, improving data retrieval response times by 40% for high-traffic operations.
- Secured the platform with CSRF protection, JWT-based authentication, encrypted user credentials, and input sanitization per OWASP standards.

Real Estate CRM & Lead Management System | [LINK](#)

November 2024 – Present

Tech Stack: Django, Bootstrap, jQuery, MySQL

- Designed and developed a comprehensive Lead Management System to capture, track, and manage real estate leads across all stages of the sales pipeline.
- Built full CRUD operations for leads, customers, and follow-up activities with optimized relational database design and foreign key constraints.
- Implemented RBAC to define and enforce permissions for administrators, sales managers, and sales executives, improving operational security.
- Optimized MySQL queries with multi-column indexing, improving lead retrieval speed and reporting performance significantly.

Research Publication & Workflow Management System | [LINK](#)

July 2024 – Present

Tech Stack: Django, Bootstrap, jQuery, MySQL

- Developed a comprehensive Research Publication & Workflow Management System to streamline journal, thesis, research paper, and book publication processes end-to-end.
- Built modules for enquiry management, lead tracking, customer onboarding, executive assignment, payment management, and publication workflow monitoring.
- Implemented RBAC to manage granular permissions for administrators, executives, and operational teams, ensuring secure multi-role access.
- Developed dynamic dashboards and reporting modules to track publication progress, customer activities, and operational KPIs in real time.

Homestay & Property Booking Platform | [LINK](#)

April 2025 – Present

Tech Stack: Django, Django REST framework, Bootstrap, jQuery, MySQL, PhonePe Payment Gateway

- Developing a comprehensive Property Management System to streamline property listings, tenant management, lease tracking, and maintenance workflows.
- Implementing secure authentication and authorization layers with RBAC to manage access for property managers, tenants, and administrators.

Product Based Employee Management System

October 2023 – December 2023

Tech Stack: Django, OAuth2, Bootstrap, jQuery, MySQL

- Developed a full-featured Employee Management System with CRUD operations for employee records, departments, roles, and payroll data.
- Integrated secure OAuth2 API authentication, enabling JWT-based authorization and RBAC for granular access control.
- Developed RESTful APIs to allow seamless integration with third-party HR management and payroll systems, improving organizational data flow.

Secure File Repository System

June 2023 – October 2023

Tech Stack: Django, Django REST framework, Bootstrap, jQuery, MySQL

- Designed and developed a secure file repository system enabling users to create folder hierarchies, upload, organize, and share files with permission-based access.
- Optimized MySQL database with indexing and foreign key constraints, improving file query execution speed by 35% for large data sets.
- Refactored the project from Django REST framework, improving raw performance and providing greater flexibility for custom business logic.

Journal Management System

February 2023 – May 2023

Tech Stack: Django, Bootstrap, MySQL

- Designed and developed a book-based journal management system enabling users to create, manage, and track reading journals and academic entries efficiently.
- Built a dynamic, mobile-responsive UI using Bootstrap, ensuring consistent cross-device experience and intuitive navigation.
- Analyzed academic performance data to identify key influencing factors, providing actionable insights through structured reporting modules.

EDUCATION

Anna University Dindigul, India

September 2018 – March 2022

Bachelor of Engineering | GPA: 8.59 / 10.0

CERTIFICATIONS | [LINK](#)

- Python Full Stack Developer Certification — Gained expertise in Python, Django, REST APIs, frontend technologies (HTML, CSS, JavaScript), and relational database management (MySQL, PostgreSQL).